

Solution Requirements

Date	28 September 2026
Team ID	6ab9f220c23107aaaa73f984
Project Name	ComicCraft AI - AI comic story creator

Functional Requirements

ID	Requirement	Priority
FR-1	User can enter a story idea, main character and setting	High
FR-2	User can pick a story tone (Funny, Adventurous, Dark, Heartwarming, Sci-Fi, Mystery, Epic)	High
FR-3	User can pick an art style (Classic Comic Book, Manga, Watercolor, Pixel Art, Noir, Cartoon, Retro Pop Art)	High
FR-4	System generates a comic title and exactly 5 connected panels via Gemini	High
FR-5	Each panel has heading, narration, dialogue, caption and an image prompt	High
FR-6	System generates one image per panel with Stable Diffusion XL	High
FR-7	Dialogue is rendered as a speech bubble over the image by the web page	Medium
FR-8	User can redraw an individual panel	Medium
FR-9	User can export the finished comic as a PDF	High
FR-10	A placeholder panel is used when image generation fails	Medium

Non-Functional Requirements

ID	Category	Requirement
NFR-1	Security	API keys are stored in backend .env and never sent to the browser; .env is excluded from Git
NFR-2	Usability	Single-page form with progress feedback and clear error messages
NFR-3	Responsiveness	Layout adapts to desktop and smaller screens
NFR-4	Reliability	One failed panel must not stop the rest of the comic
NFR-5	Maintainability	Separate modules for routes, models, script generation, images and export
NFR-6	Portability	Runs locally on Python 3.10+ with no Node.js build step
NFR-7	Performance	Script returned in seconds; image generation is the slowest step and is done per panel